

EE2703 APL Assignment 7 :

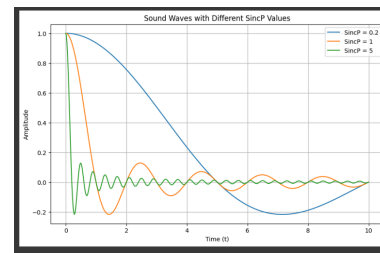
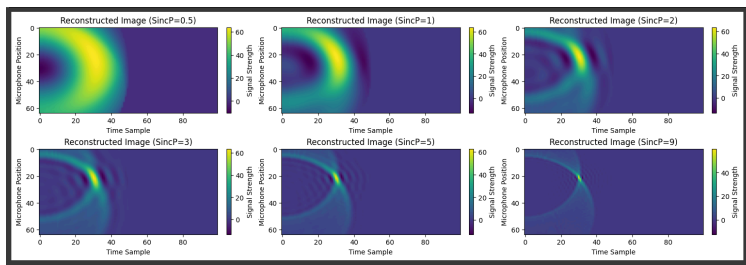
Localisation of Sound Waves:

ANSWERS TO QUESTIONS:

Question 1 : SINC Generation: We can generate these pulses by changing the value of SincP, the parameter determining the width of the Sinc wave (the sound wave). The higher the value of sincp, the more narrower the wave gets.

Coming to the final reconstructed image:

(code for plot generation is after DAS implementation)



As we can see, the obstacle is being located more precisely as we increase the value of sincP, this is because the narrower the Sinc function allows it to have a strong response over a small precise range and this will help the DAS algo to better separate delays that are close together.

Question 2: Does it make sense to reconstruct up..... here?

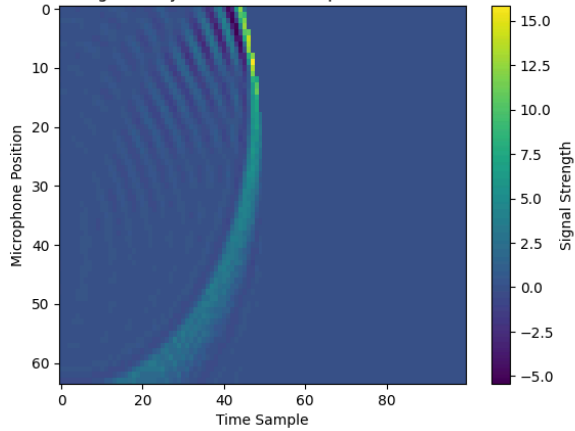
We don't have to reconstruct all N_{samples} , because the maximum time window should be long enough to capture the entirety of the signal after applying these delays. However, if the delays are relatively short compared to N_{samp} , we can use $0.5 * N_{\text{samp}}$, also Since the signal of interest will often only occupy a portion of the available time samples, reconstructing the full N_{samp} could include unnecessary data.

Questions (ordered as per the final set of questions):

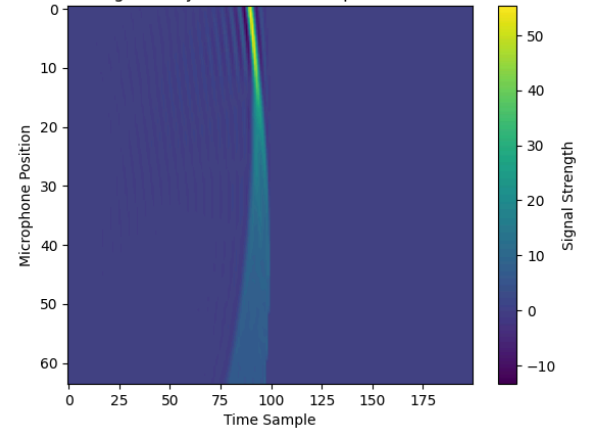
1. Here the microphone array seems to shift from '-3.15 to 3.15' to 0 to 64. centered around 32. Hence the line $y = -1$ translates to $y = 22$. Coming to X coordinate, the value 3 becomes 30 as the wave from source to obstacle then to all mics travels a distance of 30 before it is sampled there by all mics.

2. The maximum y coordinate is 3.15, where the last microphone is located and maximum x coordinate is close to 5, if the number of samples, $N_{\text{samp}} = 100$, close to 10 if $N_{\text{samp}} = 200$. This is why we only need $N_{\text{samp}} * 0.5$.

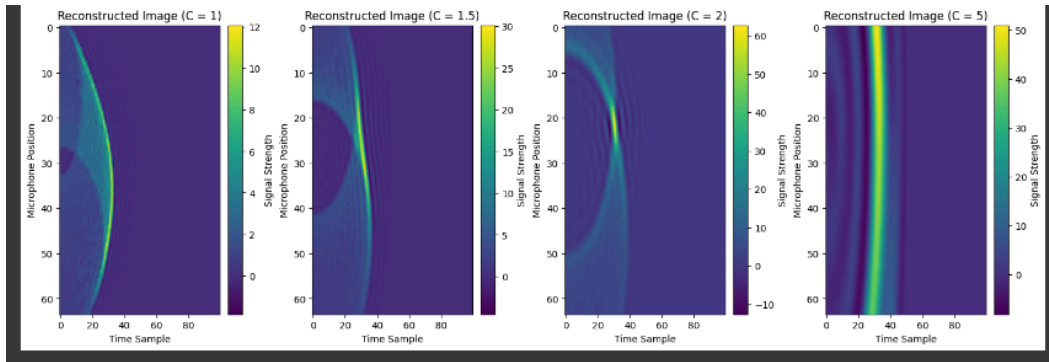
Reconstructed Image (Delay-and-Sum), Nsamp = 100, obstacle is at (4.5,-3)



Reconstructed Image (Delay-and-Sum), Nsamp = 200, obstacle is at (9,-3)

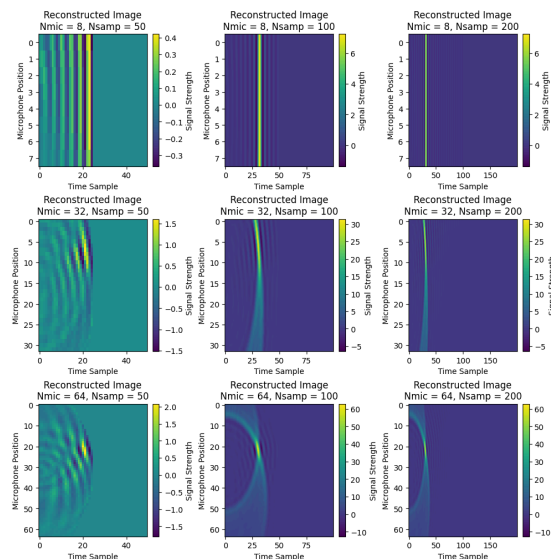


3. With more time delay between each microphone receiving the wavefront, the wave is captured more distinctly at each microphone position, leading to a better-defined pattern in the reconstructed image.



As C decreases, it means speed of sound decreases, so the delay increases and samp_time or sampling interval increases, as the wave front moves slowly it spreads out more and this wide spread is because a slower propagation speed increases the time it takes for the wave to reach different microphones, causing a more gradual, elongated pattern.

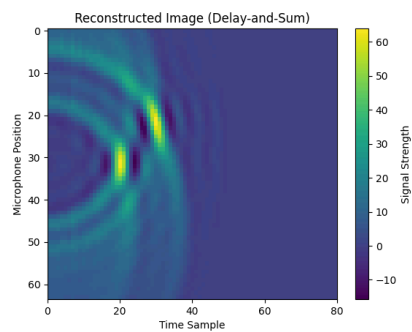
4.



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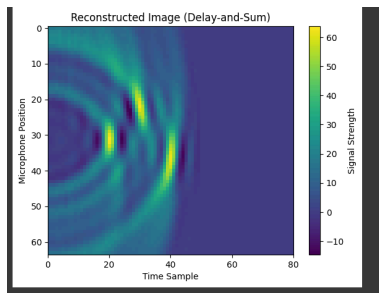
Observations: As Nmics increases, The algo locates the object more precisely and as Nsamp increases, a larger sample space is also captured.

For text files : rx2: Final reconstructed image



There are only 2 obstacles present in rx 2.

Rx3: final image:



Observation: There are 3 obstacles present